

When does fusing hand-crafted spectral knowledge with learned representations pay?

A controlled, cost-normalized benchmark: 14 datasets $\times$ 7 backbones $\times$ 500 $\rightarrow$ 1.28M images — and its organizing law

1 The fusion



$\lambda: 1 \rightarrow 0$

fixed Gabor
energy bank

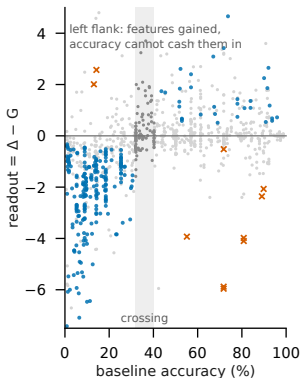
aux head
(training only)

+2% training compute

+0 at inference · identical deployed net

2 The law

$\Delta = G + \text{readout}(\text{base}) \cdot \text{sign correct in 268 of 278 testable cells}$



3 When it pays

STACK

prior + augmentation
different currencies
 $\times 1.4$ – 2.4 gain

SUBSTITUTE

prior vs. SimCLR (2 $\times$)
same currency
combo $\leq$ best single

TAX

prior on ImageNet init
overwrites mature features
–17 pts, feature-side

Headline: small ViTs, modern recipe

+13.0 (ViT-S), +26.0 (ViT-B) at 224 px
+3.2 at 1.28M images — every CNN neutral